R107/B_connected_paths | Explicit paths and actual latency

Complete standalone model including graph/path construction; no equal-FLOPs claim.

Problem	Instances	Milliseconds / instance
TSP	128	1.050937
CVRP	128	1.261312
ATSP	128	0.136482
OVRP	128	0.156624
VRPB	128	0.156675
VRPL	128	0.156738
VRPTW	128	0.157029
OVRPTW	128	0.156544
OVRPB	128	0.156752
OVRPL	128	0.156587
VRPBL	128	0.156756
VRPBTW	128	0.156543
VRPLTW	128	0.156744
OVRPBL	128	0.156806
OVRPBTW	128	0.156667
OVRPLTW	128	0.156658
VRPBLTW	128	0.156623
OVRPBLTW	128	0.156904

Total trainable parameters: 1,892,233; joint connected short-path evidence

Optional R94B: not_requested

R107 adds an explicit short-path module to the fresh R84/R88B full18 selector. A_edge_factorized uses factorized evidence; B_connected_paths uses joint connected-path computation. This is not a MoE FFN or learned expert-routing experiment. Same complete fresh tensors and total parameter counts require actual factory/initialization proof; they do not imply equal active functional freedom or equal FLOPs. Actual counts and path specification come from the locked model, never estimates. Factorization is conditional on each round incoming node states, not global selector additivity. A retains trunk interactions; B also receives explicit composed fields, so this is not an isolated topology-only causal comparison. Original R88B physical TSP node0 role, methodID+34, geometry/distance, score MLP/maximin and original CE/comparison/risk/two-forward RDropObjective remain; 57,600 updates, logical batch1024. No donor, teacher, context labels, new targets, calibration, quota or auxiliary loss. R89A AdamW, R89B Muon and R84A are saved historical references, not rerun or newly parameter/FLOP-matched controls; R94B is optional, never a prerequisite. Fixed36 rescues, losses, false picks and aligned net costs/regret are separate from choice frequency. Native/FP64 ind and legal pools; TRAIN-fitted SBS; ALL task-equal. TEST is development-exposed; one whole-model full18 VAL checkpoint for full TRAIN/VAL/TEST/LIB; no TEST/LIB reselection. Actual standalone GPU-resident-input timing includes complete model graph/path construction and computation, excludes dataset I/O, CPU transfer and solvers, and is not end-to-end dataset latency. No routing-summary requirement. No online W&B equality/availability is asserted.

R107/B_connected_paths | Same selected model: full TRAIN

Diagnostic only; TRAIN was never used to select the checkpoint; full TRAIN-fitted SBS.

Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
ALL	71.3094	66.0332	11.388904	11.400702	11.345073	-0.2000	+0.3849	0.3932
TSP	46.0593	32.8617	8.004108	8.023959	7.970215	-0.2474	+0.4252	0.4197
CVRP	47.5852	18.1284	18.038150	18.151987	17.946816	-0.6271	+0.5089	0.6702
ATSP	76.7383	46.9210	1.640090	1.678679	1.633938	-2.2987	+0.3765	0.3802
OVRP	41.1383	34.2420	7.443653	7.453531	7.397676	-0.1325	+0.6215	0.6337
VRPB	80.6198	80.2765	9.355547	9.357264	9.319342	-0.0183	+0.3885	0.3964
VRPL	37.8679	31.5309	12.393138	12.400882	12.344503	-0.0625	+0.3940	0.4009
VRPTW	80.3815	80.2407	18.812581	18.813703	18.757954	-0.0060	+0.2912	0.3047
OVRPTW	83.8827	83.8025	10.118065	10.118640	10.093220	-0.0057	+0.2462	0.2414
OVRPB	88.4148	88.3580	6.362953	6.363230	6.346658	-0.0043	+0.2568	0.2643
OVRPL	42.0938	34.4716	7.443325	7.453219	7.398483	-0.1328	+0.6061	0.6166
VRPBL	79.8642	79.4593	9.401249	9.403115	9.363196	-0.0198	+0.4064	0.4076
VRPBTW	79.6284	79.4914	19.731577	19.733549	19.642712	-0.0100	+0.4524	0.4369
VRPLTW	80.5074	80.3827	18.768531	18.769959	18.714366	-0.0076	+0.2894	0.3036
OVRPBL	88.5901	88.5556	6.358833	6.358857	6.342513	-0.0004	+0.2573	0.2649
OVRPBTW	83.3111	83.2432	10.623630	10.624446	10.586225	-0.0077	+0.3533	0.3278
OVRPLTW	83.8012	83.7321	10.110748	10.111265	10.085768	-0.0051	+0.2477	0.2443
VRPBLTW	79.6864	79.5630	19.769182	19.770814	19.680364	-0.0083	+0.4513	0.4332
OVRPBLTW	83.3988	83.3370	10.624916	10.625544	10.587365	-0.0059	+0.3547	0.3305

ALL task-equal; ratios of mean costs differ from per-instance actual regret.

R107/B_connected_paths | Fixed target36 vs R89A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	125	95	121	-0.000208439
VAL	4625	6	4	6	-0.000127597
TEST	4598	3	1	4	0.000060168

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.064883	18733	17787	-0.000208439	-0.001552
VAL	0.022222	982	964	-0.000127597	-0.000328
TEST	0.075309	1024	963	0.000060168	0.000243

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R107/B_connected_paths | Fixed target36 vs R89B

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	48	711	110	0.000827887
VAL	4625	4	39	4	0.000410383
TEST	4598	4	31	5	0.000375203

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.381756	20029	25595	0.000827887	0.009515
VAL	-0.175309	1184	1326	0.000410383	0.004677
TEST	-0.112346	1225	1316	0.000375203	0.004833

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R107/B_connected_paths | Fixed target36 vs R84A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	126	62	124	-0.000256096
VAL	4625	6	0	7	0.000093189
TEST	4598	6	1	6	0.000002611

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.065501	18329	17374	-0.000256096	-0.002049
VAL	-0.012346	980	990	0.000093189	0.001859
TEST	0.071605	996	938	0.000002611	-0.000368

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.